

Sprint 1 Handover Notes

Config-Driven RL for Just-in-Time Adaptive Interventions

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July 2026

1 Problem Definition

Physical inactivity is a leading global health risk. Just-in-Time Adaptive Interventions (JITAI) deliver personalised behaviour-change prompts via wearable devices at moments of opportunity [1]. Designing effective JITAI policies requires rapid iteration, but real-world trials are slow, expensive, and cannot explore counterfactual policies.

We formulate the problem as an MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$:

- $\mathcal{S}$: 36-state factored space (step_bin $\in \{\text{inactive, moderate, active}\}$, sleep $\in \{\text{good, poor}\}$, day_of_week $\in \{\text{weekday, weekend}\}$, burden $\in \{\text{low, medium, high}\}$)
- $\mathcal{A}$: 4 actions (idle, movement_suggestion, goal_reminder, journal)
- $\mathcal{R} = \alpha \cdot f(\text{step_bin}) + (1 - \alpha) \cdot g(\text{sleep}) - \lambda \cdot \mathbb{I}[\text{action} \neq \text{idle}]$
- 5 decision points per day, 90-day default episode, 50 seeds per agent

2 Existing Work

HeartSteps V1/V2 (Klasnja et al., 2019 [1]; Klasnja et al., 2022 [2]) — First RL-enabled MRT for physical activity. V1 ATE +35 steps; V2 introduced Thompson Sampling with dosage tracking and proxy value function for delayed effects.

PEARL (Lee et al., 2025 [3]) — Largest RL-for-PA RCT (n=13,463, 4-arm). COM-B informed 155-nudge bank. RL +296 steps vs control (p=0.0002) at 1 month, sustained at 2 months.

StepCountJITAI (Karine & Marlin, 2024 [4]) — Simulator for PA interventions, DQN $\approx 3,000$ return vs TS $\approx 1,500$.

Bandit theory Russo & Van Roy (2018) [5] establish TS regret bound $O(\sqrt{dT \log T})$. Trella et al. (2022) [6] provide design principles for RL in digital health.

LLM-based simulation Park et al. (2023) [7] demonstrate generative agents; Wang et al. (2024) use LLMs to simulate patient responses; Lu et al. (2025) benchmark LLM agents for multi-turn human behaviour.

Data-Driven Simulators & Burden Wang et al. (2021) [8] developed behavioral simulators incorporating hard frequency constraints to mitigate over-notification and habituation.

User Heterogeneity Tomkins et al. (2021) [9] introduced IntelligentPooling to balance personalization with group-level data sharing via mixed-effects Thompson Sampling. Ghosh et al. (2024) [10] extended this with reBandit (Empirical Bayes + informative priors).

Count-Based Bandits Liu et al. (2025) [11] extended TS to Poisson/NB/ZIP/ZINB models, reducing variance for skewed proximal health outcomes.

Power-Constrained Exploration Yao et al. (2021) [12] established meta-algorithms for probability clipping ($\pi_{min} \leq P(A|S) \leq \pi_{max}$) to protect post-trial statistical power in MRTs.

3 Our Approach

Two pillars: (1) config-driven framework defining entire MDPs in YAML — no source code changes between experiments; (2) LLM-bootstrapped transition models that generate plausible patient behaviour from persona descriptions.

Config-Driven Framework. State spaces, action sets, transition models, reward functions, and agent selection are defined declaratively. Three transition types: rule_based (hand-curated), random (Dirichlet baseline), and bootstrap

(LLM-generated). Expression-based rewards via safe AST parser. Seven agents implemented with contextual variants, all pluggable by name in YAML.

Synthetic LLM Patients. Transition probability tables generated by prompting LLMs (DeepSeek V4 Flash) to sample next-state distributions given a persona description, current state, and action. Pipeline: prompt generation → concurrent API calling (litellm, 200 workers) → JSON parsing → probability table aggregation. 22,320 prompts per persona (full factorial, N=10 per cell), 5 distinct behavioural personas, total cost \$0.27. 42 JSON transition tables, comprehensive analysis (parse rates, cell consistency, burden interactions, step histograms), 85 visualization figures.

4 What We Have Done

MVP Simulator. A 2-state (sedentary/active), 2-action MDP with hand-curated transition rules. 285 hyperparameter combinations swept across epsilon-greedy, decaying epsilon-greedy, UCB, and Thompson Sampling variants. Full regression test suite with golden fixtures. Demonstrates the framework pipeline end-to-end.

LLM Bootstrap. Generated 42 JSON transition tables from 22,320 LLM calls (5 personas + 2 initial model comparisons). Each persona produces distinct, behaviourally-plausible transition dynamics validated through parse rates, output distributions, cell consistency, burden interaction analysis, and step count histograms. A 36-paper literature review across 8 dimensions supports the LLM-as-transition-model methodology.

Experiments. Both MDP types run through the same YAML schema, agent library, and evaluation runner. Benchmark matrix: 7 persona configurations × 4 variant types × 7 agents × 50 seeds = 9,800 simulation runs. HeartSteps V2 reproduced faithfully (Bayesian linear regression, dosage tracking, proxy value function) as a validation agent — 4 implementation bugs independently discovered during reproduction.

5 Initial Results

Transition validation. The LLM-generated personas produce meaningfully differentiated behaviour. Burden degrades movement suggestion effectiveness: as notification burden rises from low to high, transition probabilities to higher step bins decrease systematically. Persona contrast is pronounced: identical actions applied to different personas yield fundamentally different transition probability matrices (e.g., stable maintainers retain active states; resistant users drop to inactive regardless of intervention).

Agent benchmarks. HeartSteps V2 achieves 97–100% of the optimal DP bound across all 7 persona/initial-model configurations, confirming the simulator is well-formed and can reproduce known results. Epsilon-greedy lags 10–40pp behind. Key findings:

Persona	Optimal DP	HeartSteps (% opt)	Best Bandit (% opt)
Stable Maintainer	441.8	440.8 (99.8%)	402.5 (91.1%)
Base	379.7	379.1 (99.9%)	327.8 (86.3%)
Goal-driven	414.1	414.5 (100.1%)	323.0 (78.0%)
Social Responder	313.5	306.8 (97.9%)	140.1 (44.7%)
Initial DeepSeek	381.6	373.6 (97.9%)	297.5 (78.0%)
Initial GLM 5.2	374.6	374.0 (99.8%)	326.8 (87.2%)
Resistant	214.6	163.5 (76.2%)	151.7 (70.7%)

Observations. (1) Persona effects are large — Stable Maintainer (441) vs Resistant (215). (2) Epsilon-greedy is the best bandit but lags substantially behind HeartSteps. (3) HeartSteps selects idle ≈62% of the time in the base persona — personalisation is partly about knowing when *not* to intervene. (4) Time-of-day features and step activity levels are the most predictive state variables.

6 What is Next

1. **Sensor Foundation Models.** Wearable FM POC (2,754 lines, feature branch): NormWear integration (768-dim embeddings from PPG/accelerometer), FM-guided action space (15 interventions, 3 archetypes), Thompson Sampling with FM context keys. 6 publishable research gaps identified across 12 FM models.
2. **Additional LLM Backends.** Extend bootstrapping to GLM 5.2, Qwen, Claude. Multi-turn persona dialogues for richer, temporally-aware transition models. Cost-benefit analysis across providers.
3. **Better Prompting.** COM-B behaviour-change taxonomy integration. Structured JSON schema enforcement via constrained decoding. Chain-of-thought reasoning for transition probability generation.
4. **Synthetic Clinical Results.** Calibrate LLM transitions to real-world effect sizes (PEARL: +200–300 steps). Integrate All of Us (n=59K, Fitbit) and UK Biobank (n=100K, accelerometer) for learned transition models.
5. **Algorithmic advances.** Extend beyond bandits to DQN/policy-gradient for temporal credit assignment. Power-constrained exploration [12] and count-based likelihoods [11] for future MRT deployment.

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